

Unit: Unit 6: Systems of Linear Equations and Inequalities

Name: _____

Topic: Applications of Systems of Inequalities

Date: _____

Foundation (Jalapeno)

Open Ended Questions (FRQ)

Q9. A car rental company has 50 cars and 30 trucks to rent. They charge 40 per day for cars and 60 per day for trucks. If x is the number of cars rented per day and y is the number of trucks rented per day, and the total daily revenue is 3000, write a system of inequalities that represents this situation and find the feasible region.

Q10. A factory produces two products, A and B , using two machines, X and Y . Product A requires 2 hours on machine X and 1 hour on machine Y , and product B requires 1 hour on machine X and 2 hours on machine Y . Machine X is available for 100 hours per week and machine Y is available for 80 hours per week. If x units of product A and y units of product B are produced per week, write a system of inequalities to represent the constraints and find the feasible region.

Chef's Tips

- Make sure to read the questions carefully
- Check your calculations
- Use the correct units